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CASE PORTFOLIO: CLOUD ARCHITECTURE & INFRASTRUCTURE SOLUTIONS

2008 ~ 2026

1. CLOUD STRATEGY AND DATA PLATFORM EVOLUTION

CASE #1 STRUCTURAL TRANSITION FROM DATA MESH TO MANAGED DATA PLATFORM

Grupo Globo

CONTEXT (CHALLENGE)

Redesigning and unifying the cloud infrastructure strategy to curb compute costs and simplify engineering cognitive load.

SITUATION

The decentralized Data Mesh architecture generated technical silos, massive storage redundancy, and prohibitive multi-cloud licensing and processing costs.

TASK

Migrate and consolidate the data infrastructure into a 'Managed Platform' model without violating the business autonomy of the squads.

ACTION

Designed a centralized architecture unifying repositories and abstracting corporate database connections via microservices layers.

RESULT

Severe optimization of active computational assets, a drastic reduction in engineers' cognitive load, and stabilization of license costs.

BUSINESS METRICS

- Substantial reduction in team cognitive load during onboarding
- Optimization and stabilization of overall software licensing costs

TECHNICAL ARCHITECTURE

- Managed Public Cloud Environments (AWS / GCP)
- Decoupled data abstraction layers based on robust microservices

GOVERNANCE

- Centralized corporate infrastructure policies and standards
- Federated data architecture compliance and validation rituals

CASE #2 STANDARDIZATION OF CLOUD ENVIRONMENT PROVISIONING VIA GOLDEN PATHS

Grupo Globo

CONTEXT (CHALLENGE)

Automation and scalable governance of access to cloud resources for a massive ecosystem of agile squads.

SITUATION

Engineering teams suffered from weeks of technical delays waiting for manual provisioning of cloud infrastructure and access privileges.

TASK

Design automated self-service paths (Golden Paths) to accelerate the Time-to-Value of new products.

ACTION

Created standardized Infrastructure as Code (IaC) blueprints enabling secure self-provisioning of data environments.

RESULT

The infrastructure setup and access governance timeline for over 100 squads was drastically reduced from days to seconds.

BUSINESS METRICS

- Reduction of analytics setup time (Time-to-Value) from days to seconds
- Absolute standardization of access security for more than 100 squads

TECHNICAL ARCHITECTURE

- Data Platform as a Service (DPaaS)
- Automated cloud infrastructure provisioning workflows (Golden Paths)

GOVERNANCE

- Automated least privilege policies (Least Privilege)
- Centralized monitoring and automated auditing of privileged access

CASE #3 UNIFICATION AND CONSOLIDATION OF ARCHITECTURES FOR ENTERPRISE SAP/BI SYSTEMS

Pilkington Blindex / CI&T

CONTEXT (CHALLENGE)

Structural technical integration between complex transactional backends and enterprise analytical layers.

SITUATION

Severe data extraction inconsistencies and storage I/O bottlenecks between the SAP ERP and BI repositories generated analytical failures.

TASK

Design and coordinate the unifications of data pipelines to ensure absolute referential integrity.

ACTION

Led data cleansing efforts, refactored logical schemas, and restructured ETL flows for high-throughput processing.

RESULT

Achieved 100% reliability in analytical data with severe processing gains, generating recurring savings of R$ 1.2 million/year.

BUSINESS METRICS

- Audited recurring savings of R$ 1.2 million per year
- Achieved 100% reliability and integrity across analytical bases

TECHNICAL ARCHITECTURE

- Integrated SAP ERP enterprise systems architecture
- High-performance ETL pipelines for Business Intelligence

GOVERNANCE

- Formal rituals of a dedicated data sanity and data quality squad
- Strict key validation and database integrity checks

CASE #4

MULTIDISCIPLINARY ARCHITECTURAL DESIGN FOR SINGLE PLATFORM MODERNIZATION

Gafisa / CI&T

CONTEXT (CHALLENGE)

Technical consolidation of scattered legacy ecosystems into a modern, unified infrastructure.

SITUATION

Three legacy systems operated in a disconnected manner with serious synchronization issues, misaligning engineering and data teams.

TASK

Unify legacy systems into a single operations platform, mitigating technical and architectural risks.

ACTION

Designed the blueprint for the unified platform, established robust integrations via APIs, and coordinated the technological transition with tech leads.

RESULT

Modernization completed with a drastic reduction in Time-to-Market for new features and USD 1.2 million in operational savings in the first year.

BUSINESS METRICS

- 35% reduction in Time-to-Market for new features
- Proven operational savings of USD 1.2 million in the first year

TECHNICAL ARCHITECTURE

- Unified enterprise operations platform
- Modern legacy system integration layers via robust APIs

GOVERNANCE

- Cross-functional governance matrix integrated with tech leads
- Technical scope alignment driven by shared OKRs

2. AI ARCHITECTURE AND REGULATED MLOPS BLUEPRINTS

CASE #5 SECURE MLOPS BLUEPRINT FOR GENERATIVE ARTIFICIAL INTELLIGENCE DEPLOYMENT

PwC Brasil

CONTEXT (CHALLENGE)

Designing a shielded cloud infrastructure for AI processing in highly regulated environments.

SITUATION

Advanced Natural Language Processing (NLP) and Generative AI models faced deployment locks due to audit risk assessments.

TASK

Design a structural MLOps architecture that meets rigid compliance and security criteria.

ACTION

Projected secure pipelines featuring container isolation, automatic encryption, and data anonymization routines.

RESULT

The blueprint was 100% approved by risk auditing, safely compressing document analysis cycle times by 70%.

BUSINESS METRICS

- 70% reduction in operational cycle time for document reviews
- Zero security or non-compliance findings during risk audit

TECHNICAL ARCHITECTURE

- Secure MLOps platform architecture blueprint focused on advanced protection
- Large Language Models (LLMs) and Generative AI frameworks

GOVERNANCE

- Strict alignment with data privacy laws (LGPD/GDPR) and ISO standards
- Multi-disciplinary sign-off gates (Engineering, Risk, and Audit)

CASE #6 CRITICAL PATH MANAGEMENT (CPM) FOR HIGH-PERFORMANCE COMPUTE CLUSTER PROVISIONING

PwC Brasil

CONTEXT (CHALLENGE)

Optimizing infrastructure schedules to neutralize technical bottlenecks and third-party deadlines.

SITUATION

Critical dependencies on public cloud GPU instance provisioning and legal data privacy checks threatened to blow past project deadlines.

TASK

Ensure strict adherence to milestones and mitigate delay risks from external dependencies.

ACTION

Applied Critical Path Method (CPM) modeling to the cloud schedule, accelerated timelines, and projected an open fallback architecture.

RESULT

Integration executed with absolute success by the code freeze date, securing on-time delivery and eliminating penalties.

BUSINESS METRICS

- 100% On-Time Delivery performance within the final deadline
- Total elimination of contractual fine risks due to delays

TECHNICAL ARCHITECTURE

- High-performance cloud compute clusters (GPUs)
- Open architecture based on decoupled microservices for technical fallback

GOVERNANCE

- Schedule tracking based on the Critical Path Method (CPM)
- Proactive acceleration of procurement and legal compliance alignment loops

CASE #7

FINANCIAL BASELINE AND INFRASTRUCTURE SCOPE FOR MACHINE LEARNING MODELS

Grupo Boticário / Accenture

CONTEXT (CHALLENGE)

Containing scope creep and hidden cloud processing costs during incremental algorithm training.

SITUATION

The highly iterative nature of refining predictive search algorithms posed a serious risk of cloud budget overruns.

TASK

Implement technical governance mechanisms to shield the project against computational waste.

ACTION

Designed an architecture with strict workload thresholds and conditioned cloud consumption on rigid Definition of Done (DoD) criteria.

RESULT

Overall cloud budget variance was contained below 2%, fully preserving the project's targeted 22% profit margin.

BUSINESS METRICS

- Final cloud budget variance kept strictly under 2%
- Full preservation of the project services profit margin at 22%

TECHNICAL ARCHITECTURE

- Algorithm training pipelines with rigid processing thresholds
- Data accuracy telemetry and infrastructure consumption tracking

GOVERNANCE

- Budget governance committee with weekly Burn Rate analysis
- Definition of Done (DoD) criteria tightly coupled to AI efficiency targets

CASE #8 CMS-AI ARCHITECTURE DECOUPLING VIA VERTICAL SLICING

Grupo Globo / Accenture

CONTEXT (CHALLENGE)

Mitigating integration risks in complex cloud architectures and mutual system dependencies.

SITUATION

High technical integration risk between enterprise content management systems and predictive artificial intelligence models.

TASK

Structure an architectural approach that enables early and incremental validation of the integrated environment.

ACTION

Enforced strict vertical scope slicing (Vertical Slicing), isolated components using decoupled microservices via APIs, and validated an MVP by Sprint 3.

RESULT

Reduced technical integration risks by 40% early in the schedule, enabling stable continuous deployments every two weeks.

BUSINESS METRICS

- 40% reduction in system technical integration risks
- Guaranteed stable incremental deployments every two weeks

TECHNICAL ARCHITECTURE

- Adobe Experience Manager (AEM) CMS integration with AI models
- Decoupled microservices design based on robust APIs

GOVERNANCE

- Technical strategy based on Vertical Slicing
- Technical synchronization via scaled agile frameworks (Scrum of Scrums)

3. CLOUD FINOPS AND INFRASTRUCTURE BUDGET OPTIMIZATION

CASE #9 CONTROLLED DOWNGRADE AND CLOUD COST OPTIMIZATION VIA FINOPS

Accenture (SolutionsIQ) / Accenture

CONTEXT (CHALLENGE)

Applying cloud financial governance (FinOps) to control runaway public cloud spend.

SITUATION

Corporate servers and instances operated superdimensioned (oversized), creating massive computational and budgetary waste.

TASK

Optimize the cloud infrastructure footprint and reduce recurring costs without compromising system stability.

ACTION

Implemented systematic monitoring of compute capacity and executed technical redimensioning (Right-sizing) of instances.

RESULT

Delivered an immediate and recurring 28% savings in the global cloud infrastructure budget.

BUSINESS METRICS

- 28% cost reduction in public cloud invoicing
- Significant increase in infrastructure asset investment ROI

TECHNICAL ARCHITECTURE

- Cloud resource profiling and right-sizing models
- Cloud cost monitoring and telemetry systems

GOVERNANCE

- Cloud financial governance principles and practices (FinOps)
- Periodic cost-efficiency architectural reviews

CASE #10

FINANCIAL AND TECHNICAL CONSOLIDATION OF CLOUD LICENSING PER CORE

insi (Meta IT)

CONTEXT (CHALLENGE)

Auditing cloud resource utilization to mitigate contractual and financial risks of proprietary software.

SITUATION

Instance sprawl generated waste in maximizing high-cost corporate database software licenses billed per physical core.

TASK

Centralize infrastructure and adapt logical architecture to fit strictly within contracted license thresholds.

ACTION

Conducted a technical resource audit, deactivated idle components, and consolidated databases onto high-density dedicated hosts.

RESULT

Guaranteed absolute technical compliance with the vendor, eliminating compliance penalty risks by 100%.

BUSINESS METRICS

- Complete elimination of heavy non-compliance fine risks
- Maximum optimization of expensive corporate software license usage

TECHNICAL ARCHITECTURE

- Cloud hardware physical core limitation matrix
- Centralized multi-instance server topology

GOVERNANCE

- Strict corporate resource utilization audit routines
- Formal control policies for premium feature activation

CASE #11

HYBRID STORAGE STRATEGY FOR STORAGE GROWTH OPTIMIZATION

Grupo Globo

CONTEXT (CHALLENGE)

Smart management of voluminous data lifecycles to avoid inflated costs on premium storage media.

SITUATION

The exponential growth of Terabytes of logs and old history data on ultra-expensive production SSD arrays suffocated the IT budget.

TASK

Design an automated lifecycle and retention strategy to shift historical records to low-cost storage tiers.

ACTION

Projected automated data tiering routines (Storage Tiering Layouts) and implemented cloud cold storage lifecycle policies.

RESULT

Immediate reclamation of 40% usable space on main premium production SSD arrays, significantly delaying Capex spend.

BUSINESS METRICS

- Immediate 40% capacity reclamation on premium production storage media
- Significant postponement of storage infrastructure Capex investments

TECHNICAL ARCHITECTURE

- Hybrid storage tiering layouts
- Cloud cold storage lifecycle policies integrated with low-cost media

GOVERNANCE

- Formal corporate retention and historical data lifecycle policies
- Regulatory validation and sign-off on data purging and discarding rules with the business

CASE #12 TRACEABILITY AND ACCOUNTING SEGREGATION OF INFRASTRUCTURE FOR TAX/COMPLIANCE AUDITS

Grupo Globo

CONTEXT (CHALLENGE)

Automating cloud engineering cost controls by integrating infrastructure metadata with ledger account rules.

SITUATION

The lack of clear distinction between infrastructure for asset creation (Capex) and operational maintenance (Opex) generated serious fiscal compliance risks.

TASK

Structure an un-gameable cloud infrastructure tagging matrix to automatically segregate costs.

ACTION

Tied cloud provisioning metadata directly to CI/CD pipelines and created automated accounting dashboard connectors.

RESULT

Brought absolute financial audit clarity to the project, with the consolidated operation coming in 5% under the original baseline budget.

BUSINESS METRICS

- Consolidated project delivered 5% under the baseline budget
- Total mitigation of corporate tax compliance or non-conformity risks

TECHNICAL ARCHITECTURE

- Integrated cloud billing and time-tracking software layers
- Automated cost dashboarding connectors and pipelines

GOVERNANCE

- Strict corporate allocation rules separating Capex and Opex streams
- Weekly accounting ledger reviews matching engineering velocity to asset creation

4. FLOW MANAGEMENT, TELEMETRY, AND CLOUD ENGINEERING EFFICIENCY

CASE #13 WIP LIMITS AND SWARMING TO REDUCE INFRASTRUCTURE DELIVERY LEAD TIME

insi (Meta IT)

CONTEXT (CHALLENGE)

Optimizing the workflow of senior engineering teams by eliminating context-switching waste.

SITUATION

Highly senior engineering talent suffered from parallel demand overload, generating chronic bottlenecks and delayed releases.

TASK

Accelerate the release of infrastructure packages through the implementation of analytical flow management.

ACTION

Enforced rigid Work-In-Progress (WIP) limits across engineering board stages and activated daily team swarming.

RESULT

Achieved an immediate 20% compression in software delivery Lead Time with full flow stabilization.

BUSINESS METRICS

- 20% reduction in software delivery Lead Time
- Complete stabilization of team productivity and continuous throughput

TECHNICAL ARCHITECTURE

- Advanced analytical Kanban tracking systems
- Automated alerts for idle tasks and technical blockers

GOVERNANCE

- Kanban WIP limit policies and team working agreements (Kanban KMP)
- Daily blocker-resolution and swarming routines for code reviews

CASE #14

VALUE STREAM MAPPING (VSM) FOR CI/CD PIPELINE AUTOMATION

Natura / Accenture

CONTEXT (CHALLENGE)

Systematic elimination of bureaucratic steps and manual validations within the cloud deployment pipeline.

SITUATION

The enterprise production deployment pipeline contained excessive manual gates and hidden wait states, delaying Time-to-Market.

TASK

Identify systemic waste and redesign the software engineering lifecycle end-to-end.

ACTION

Facilitated Value Stream Mapping (VSM) workshops, identified redundant checkpoints, and automated quality gates via CI/CD pipelines.

RESULT

Secured a 15% reduction in overall engineering operational costs and optimized product Cycle Time.

BUSINESS METRICS

- 15% reduction in engineering operational waste
- Significant compression of delivery Cycle Time

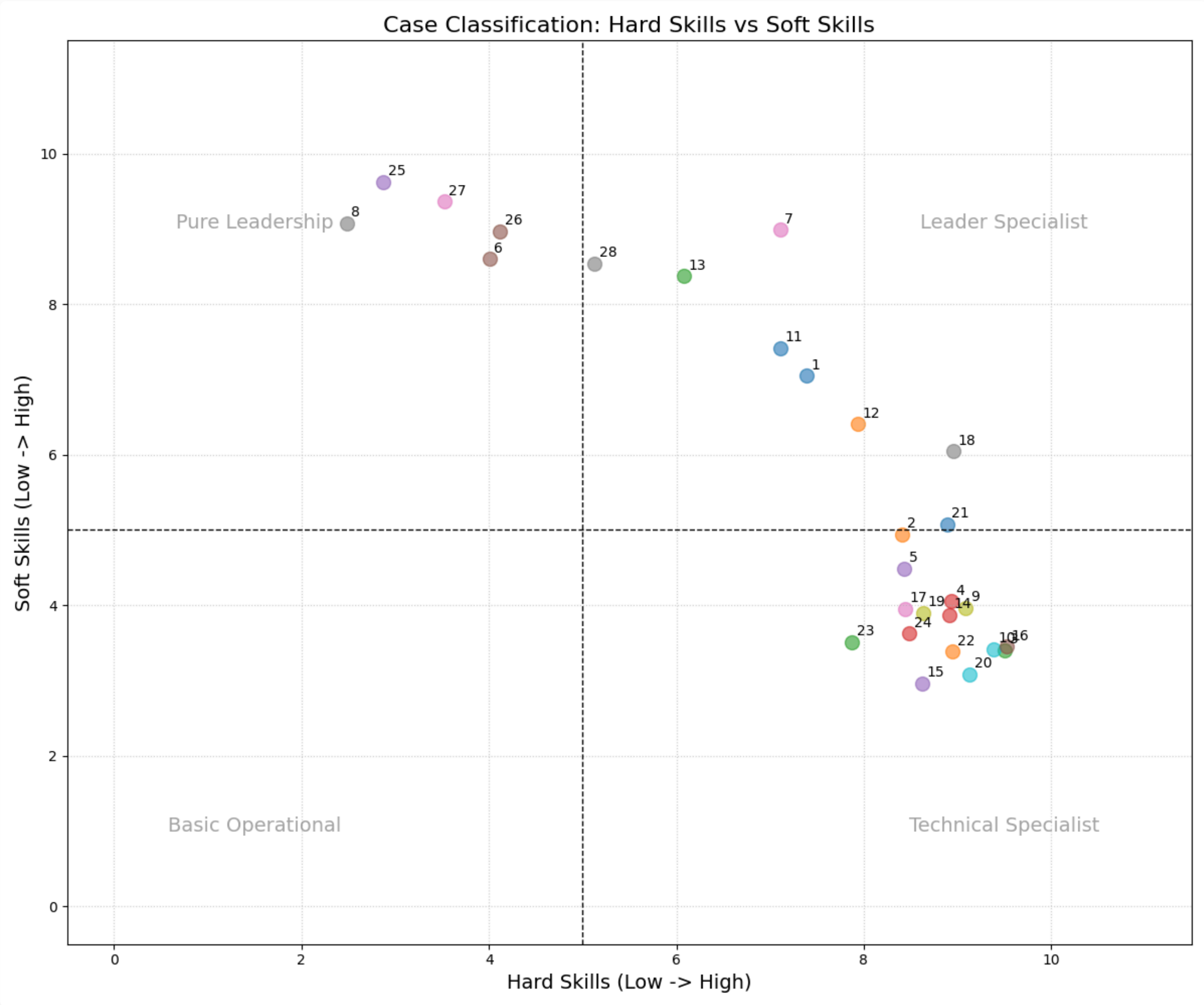
TECHNICAL ARCHITECTURE

- Automated CI/CD (Continuous Integration / Continuous Deployment) pipelines
- Analytical software operational flow mapping and modeling

GOVERNANCE

- Fully automated quality gates and release approval workflows
- Continuous improvement loops based on Lean Digital mapping models

CASE PORTFOLIO ANALYSIS: PROJECT MANAGEMENT



Visualization of the distribution of the 28 cases based on the balance between **Hard Skills** (Technical/Processes) and **Soft Skills** (Leadership/Resilience).

CASE #15 MATHEMATICAL FLOW TELEMETRY BASED ON CUMULATIVE FLOW DIAGRAMS (CFD)

Grupo Globo / Accenture

CONTEXT (CHALLENGE)

Using analytical intelligence and operational data modeling to replace subjective estimations.

SITUATION

Squad status reporting relied heavily on subjective impressions, failing to point out expanding queues or scope creep.

TASK

Implement an entirely objective technical telemetry system and empirical data-backed forecasting.

ACTION

Configured and monitored Cumulative Flow Diagrams (CFD), isolating expanding mathematical bands to diagnose bottlenecks.

RESULT

Subjectivity was entirely eliminated from scope negotiations, increasing delivery date estimation accuracy by 30%.

BUSINESS METRICS

- 30% boost in delivery date forecasting accuracy
- Early detection of expanding operational queues before delay occurs

TECHNICAL ARCHITECTURE

- Automated operational data extraction scripts via development tool APIs
- Real-time Cumulative Flow Diagram (CFD) analytical tracking dashboards

GOVERNANCE

- Data-driven scope negotiation and prioritization policies
- Weekly mathematical flow diagnostic reviews with engineering leaders

CASE #16 PYTHON/PANDAS STATISTICAL ANALYSIS FOR VELOCITY AND CLOUD ASSET ROI CORRELATION

Grupo Globo

CONTEXT (CHALLENGE)

Converting raw software engineering logs into executive business intelligence.

SITUATION

There was a complete lack of empirical visibility linking squad delivery velocity to the financial return of new digital channels.

TASK

Develop an analytical data pipeline that translates raw commit metadata into C-Level business insights.

ACTION

Wrote structured Python scripts leveraging Pandas to process operational logs and cross-reference client engagement datasets.

RESULT

Enabled lean portfolio optimization, directing funding toward high-efficiency tracks and driving a 15% productivity gain.

BUSINESS METRICS

- 15% increase in operational efficiency across prioritized areas
- Alinhamento of tech funding based on actual financial return (ROI)

TECHNICAL ARCHITECTURE

- Automated scripts in Python for log parsing and consolidation
- Analytical data pipelines for cross-database trend matching with Pandas

GOVERNANCE

- Periodic statistical auditing of engineering performance
- Data modeling architectures converting code inputs into business value metrics

⚠ 5. CLOUD INFRASTRUCTURES TURNAROUND AND CRITICAL INCIDENTS MANAGEMENT

CASE #17 NETWORK ARCHITECTURE TURNAROUND AND EMERGENCY CONNECTION POOLING OPTIMIZATION

Pernambucanas / Meta IT

CONTEXT (CHALLENGE)

Recovery of an environment facing severe technical collapse and continuous processing instability.

SITUATION

The primary transactional database crashed repeatedly in production due to severe connection leaks from the application layer.

TASK

Assume leadership of the crisis under intense business pressure and restore operational stability.

ACTION

Stood up an emergency technical War Room, applied strict pooling limits, and implemented session-killing scripts.

RESULT

Total platform stabilization achieved within hours, reducing platform downtime by 90% and safeguarding live revenue.

BUSINESS METRICS

- 90% reduction in downtime during critical system crash
- Immediate protection and stabilization of digital revenue channels

TECHNICAL ARCHITECTURE

- Advanced Connection Pooling configurations in cloud environments
- Live application activity monitoring infrastructures

GOVERNANCE

- Resolute stoic leadership in high-pressure War Rooms
- Formalized post-incident root cause analysis and corrective review with engineering

CASE #18 DW INFRASTRUCTURE RECOVERY AND POINT-IN-TIME RECOVERY (PITR) POST PHYSICAL FAILURE

Grendene / Meta IT

CONTEXT (CHALLENGE)

Catastrophic emergency rescue of data integrity following a sudden severe hardware crash.

SITUATION

A severe power surge corrupted physical blocks of the primary storage array, rendering the database inaccessible.

TASK

Recover logical data consistency and restore manufacturing operations in the shortest technical time physically possible.

ACTION

Executed deep logical diagnostic routines, extracted unaffected tables, and restored the base via log transaction backups (PITR).

RESULT

Database restored with successful recovery and zero records lost, reestablishing the operational manufacturing routine.

BUSINESS METRICS

- Zero corporate data lost during severe physical hardware crash
- Immediate return of operational capacity for interrupted manufacturing plants

TECHNICAL ARCHITECTURE

- Point-in-Time Recovery (PITR) advanced mechanisms
- Database transaction tail-log backups

GOVERNANCE

- Structured technical operational contingency frameworks
- Formal maximum severity reporting structure tailored for C-Level executives

CASE #19

ISOLATING MEMORY LEAKS IN LEGACY CLOUD CONNECTION DRIVERS

Santo Antônio Energia / CI&T

CONTEXT (CHALLENGE)

Diagnosing a sutil, cumulative intermittent fault across critical power generation monitoring data servers.

SITUATION

Every four days, the data server suffered from a complete crash due to RAM resource exhaustion, requiring constant preventive reboots.

TASK

Track down in a deep and surgical manner which logical process or query was pinning system memory clerks.

ACTION

Utilized Dynamic Management Views (DMVs), isolated internal buffers, and discovered a native bug in a legacy connection driver.

RESULT

Replaced the defective driver and adjusted the maximum memory allocation, generating continuous, uninterrupted stability.

BUSINESS METRICS

- 100% elimination of manual preventive reboots on servers
- Uninterrupted stability of the technical ecosystem monitoring energy generation

TECHNICAL ARCHITECTURE

- Dynamic Management Views (DMVs) for deep cache buffer analysis
- Memory clerk tracking frameworks

GOVERNANCE

- Technical validation and sign-off routine for third-party drivers
- Strict maximum memory threshold policies set for corporate instances

CASE #20 EMERGENCY RECOVERY OF MIGRATION PROJECT WITH SCOPE REDUCTION TO MVP

Mapfre Seguros / Accenture

CONTEXT (CHALLENGE)

Architectural intervention in distressed IT programs to restore timeline and budget baselines.

SITUATION

A complex migration project suffered from a six-month delay and an entirely blown, un-managed budget.

TASK

Reverse negative performance metrics, restructure the roadmap, and regain control of the product lifecycle.

ACTION

Conducted an infrastructure diagnostic, froze ongoing technical modifications, and cut the scope down to a lean, essential MVP.

RESULT

The migrated MVP was successfully launched within the new deadline, stabilizing budget and client satisfaction.

BUSINESS METRICS

- Financial stabilization following massive budget baseline overruns
- Complete recovery of customer confidence and Net Promoter Score (NPS)

TECHNICAL ARCHITECTURE

- Temporary architectural change freeze frameworks
- Deep codebase and infrastructure audit toolsets

GOVERNANCE

- Total reset of the project governance model
- Strict weekly checkpoint gates bound to an aggressive scope reduction plan

5. ROLLOUT OF ENTERPRISE CRITICAL AND HYBRID SYSTEMS

CASE #21 HIGH-COMPLEXITY GLOBAL STRUCTURAL MIGRATION WITHOUT PRODUCTION DOWNTIME

Deutsche Bank / CI&T

CONTEXT (CHALLENGE)

Planning and execution of a transition architecture for cross-continental financial database replication.

SITUATION

The board required the unification and mirroring of critical global repositories while maintaining fault tolerance and zero data loss.

TASK

Design a highly stable replication model at the storage block level.

ACTION

Configured synchronous and asynchronous replication in Maximum Availability mode, structuring rigid archive log validation loops.

RESULT

International financial environment shielded against downtime, enabling production switchover with zero loss of active records.

BUSINESS METRICS

- Real-time international data synchronization without losses
- Zero incidence of financial record loss or corruption during transitions

TECHNICAL ARCHITECTURE

- Advanced geographic replication mechanisms (Data Guard / AlwaysOn Availability Groups)
- Logical systems for monitoring replication sync delay

GOVERNANCE

- Automated continuous auditing of replication lag
- Compliance with international banking regulatory rules

CASE #22 ZERO-DOWNTIME MIGRATION TO MANAGED CLOUD POSTGRESQL VIA MESSAGING BUSES

Magazine Luiza / CI&T

CONTEXT (CHALLENGE)

Strategic transition from expensive proprietary database engines to managed open-source solutions in the cloud.

SITUATION

High transaction volumes generated prohibitive and suffocating per-processor-core licensing costs in the legacy model.

TASK

Migrate the data layer of the core product catalog to the public cloud (GCP) with zero live environment impact.

ACTION

Designed a real-time parallel synchronization strategy using Apache Kafka messaging buses during the migration window.

RESULT

The final cutover to the cloud was executed with absolute success and zero seconds of interruption to the user's shopping experience.

BUSINESS METRICS

- Zero real downtime generated during final cutover of production database
- Complete elimination of proprietary software license costs

TECHNICAL ARCHITECTURE

- Managed Google Cloud SQL PostgreSQL repositories
- Apache Kafka structured for parallel synchronization and concurrency

GOVERNANCE

- Automated technical validation windows for hash consistency
- Strict concurrency control and hybrid data write integrity rules

CASE #23 SYNCHRONIZATION AND SLICING OF SAP S/4HANA IN-MEMORY BACKEND ARCHITECTURE

Grupo Odebrecht / CI&T

CONTEXT (CHALLENGE)

Adapting legacy heterogeneous databases to support consolidation in un-gameable integrated ERP systems.

SITUATION

Contract and engineering data resided in obsolete legacy structures, incompatible with the technical rigor of SAP S/4HANA.

TASK

Coordinate logical mapping (Data Mapping) and execute massive data loads while ensuring 100% referential integrity.

ACTION

Projected advanced scripts to clean up orphan records, validated foreign keys, and led in-memory data loads via dedicated pipelines.

RESULT

None

BUSINESS METRICS

- Achieved 100% referential integrity across structured databases
- Enabled structural ERP go-live without data-related delays

TECHNICAL ARCHITECTURE

- SAP Hana Memory Database Backend
- Data Mapping pipelines and advanced data cleansing scripts

GOVERNANCE

- Critical Systems Transition Governance Committee
- Compliance validation of historical contracts

CASE #24 MULTI-NODE HIGH AVAILABILITY SQL SERVER ALWAYSON CLUSTER DESIGN

Mapfre Seguros / Accenture

CONTEXT (CHALLENGE)

Eliminating physical single points of failure in critical policy issuance and billing systems.

SITUATION

The database infrastructure operated on a single local server, creating high operational risk of business interruption.

TASK

Design and implement a fault-tolerant hybrid cloud architecture with instant automatic failover.

ACTION

Configured Windows Server Failover Clustering (WSFC) integrated with AlwaysOn Availability Groups with synchronous replicas.

RESULT

Disponibilidade elevated to 99.99% for the critical layer, ensuring business continuity during node failures.

BUSINESS METRICS

- Elevation of availability to 99.99% matching business SLAs
- Reduction of RTO (Recovery Time Objective) to near-zero thresholds

TECHNICAL ARCHITECTURE

- MS SQL Server Enterprise Edition
- Windows Server Failover Clustering (WSFC) & AlwaysOn

GOVERNANCE

- Scheduled cyclic test simulations of automatic failover processes
- Risk-approved structured contingency matrix for hardware failures

6. BACKUP AUTOMATION, MONITORING, AND CUSTOM SCRIPTING

CASE #25 AUTOMATED HANDS-FREE BACKUP VALIDATION VIA CUSTOM OBJECT-ORIENTED PYTHON CLASS

MLGTT Consulting

CONTEXT (CHALLENGE)

Creating object-oriented automation utilities for continuous integrity checking of storage repositories.

SITUATION

Infrastructure backups were generated daily but lacked automated validation routines against corrupted files.

TASK

Automate a continuous pipeline to perform test restores and consistency validation checks autonomously.

ACTION

Wrote a robust Python class that orchestrates local backups, transfers them via cloud APIs, and executes logical integrity checks (DBCC).

RESULT

Built an automated hands-free validation pipeline with automatic notifications via Slack, shielding DR plans.

BUSINESS METRICS

- 100% verified operational certainty of backup files integrity
- Complete elimination of daily manual effort in technical verifications

TECHNICAL ARCHITECTURE

- Structured and versioned Python automation scripts framework
- Public cloud object storage APIs (AWS S3 / Google Cloud Storage API)

GOVERNANCE

- Strict continuity plan validation policies
- Structured version control of internal utility scripts (mlgtt_utils)

CASE #26 PREDICTIVE ALERTING AND TELEMETRY ECOSYSTEM VIA SHELL SCRIPT AND GRAFANA

Claro / Accenture

CONTEXT (CHALLENGE)

Collecting and consolidating local OS hardware metrics to anticipate storage and memory exhaustion.

SITUATION

Severe incidents like low disk space or memory exhaustion brought down critical instances overnight without warning.

TASK

Develop a predictive monitoring model to alert on-call engineers before a fatal system collapse occurred.

ACTION

Wrote local Shell Scripts to collect OS telemetry, consolidated them into Prometheus repositories, and generated PagerDuty triggers in Grafana.

RESULT

Drastic reduction in surprise resource incidents, enabling operational anticipation of infrastructure issues up to 12 hours in advance.

BUSINESS METRICS

- Elimination of unexpected production stops from physical resource exhaustion
- Predictive tracking of disk limits with an anticipation window of up to 12 hours

TECHNICAL ARCHITECTURE

- Linux Shell Scripting for native OS telemetry extraction
- Prometheus / Telegraf collector engines integrated with Grafana

GOVERNANCE

- Technical severity threshold matrix defined by criticality
- Formal automatic escalation workflows for preventive alerts

CASE #27 PYTHON SCRIPT ENGINE (PILLOW) FOR HIGH-VOLUME GRAPHICAL PROCESSING AUTOMATION

MLGTT Consulting

CONTEXT (CHALLENGE)

Optimizing slow operational tasks through specialized custom code tailored for image manipulation workflows.

SITUATION

The manual routine of resizing and merging corporate logo mosaics created severe operational bottlenecks and required days of layout design work.

TASK

Build a script utility to automate batch processing, compressing layout execution times to minutes.

ACTION

Developed the 'MosaicGenerator' Python class leveraging the Pillow library and applying advanced Lanczos scaling interpolation algorithms.

RESULT

Slashed execution timeline from working days down to mere minutes, driving exponential internal productivity gains.

BUSINESS METRICS

- Drastic operational compression from working days to minutes
- Exponential increase in internal asset processing capacity

TECHNICAL ARCHITECTURE

- Structured, object-oriented, and reusable Python scripts
- Pillow graphical processing library utilizing Lanczos scaling

GOVERNANCE

- Version-controlled local utility scripts repository (mlgtt_utils)
- Corporate reuse policies for internal automation modules

CASE #28 INFRASTRUCTURE-AS-CODE (IAC) AUTOMATION VIA SOFTWARE LIFECYCLE RELEASE PIPELINES

insi (Meta IT)

CONTEXT (CHALLENGE)

Inserting logical schema modifications directly into automated CI/CD pipelines (Database DevOps).

SITUATION

Manual execution of raw SQL scripts caused frequent syntax errors and environment crashes during critical deployments.

TASK

Eliminate the human error factor by completely halting direct manual commands in production environments.

ACTION

Implemented the Flyway framework for incremental database versioning tied automatically to the Git repository.

RESULT

Full standardization of database deployments, achieving zero operational downtime from forgotten manual parameters.

BUSINESS METRICS

- Complete elimination of environment failures from incorrectly executed SQL scripts
- Significant reduction in time needed for technical patch deployment

TECHNICAL ARCHITECTURE

- Flyway Database Migrations Framework
- Automated CI/CD pipelines via GitLab CI

GOVERNANCE

- Strict corporate policies prohibiting manual schema manipulation
- Formal technical approval process for pull requests via Tech Leads